

Technical Report: Predictive Maintenance Analysis and Anomaly Detection

This report presents a comprehensive analysis of predictive maintenance using Machine Learning techniques and Anomaly Detection. Simulated temporal data was generated for multiple equipment over 40 time steps to predict failures and detect anomalies in their evolution. Multiple models were trained and their performance was evaluated using standard metrics and detailed visualizations.

1. Exploratory Data Analysis (EDA)

Figure 1: Failure vs No Failure Distribution

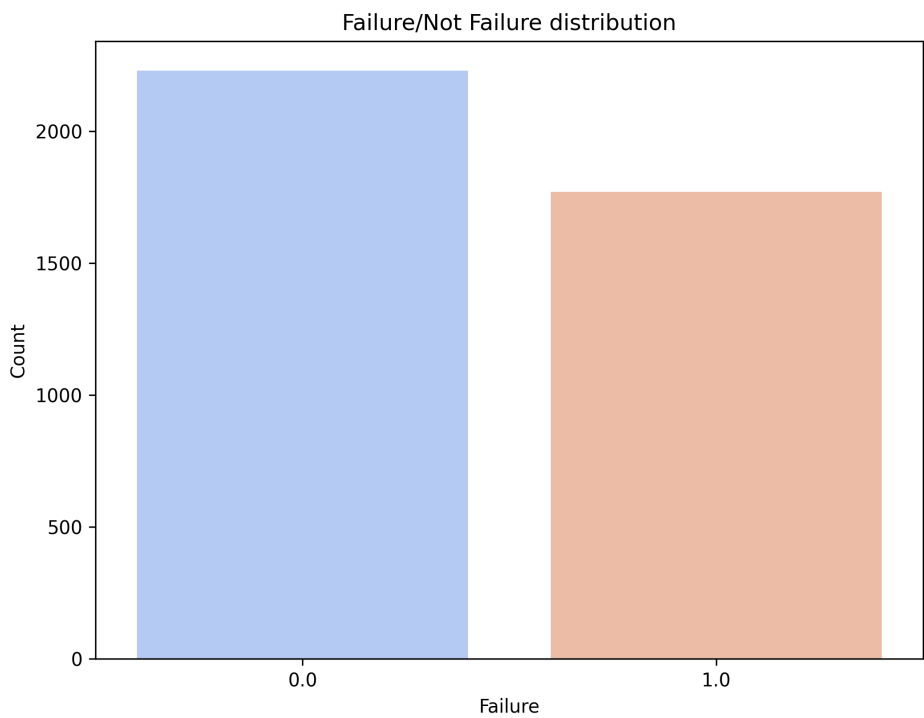


Figure 2: Simulated Data Correlation Matrix

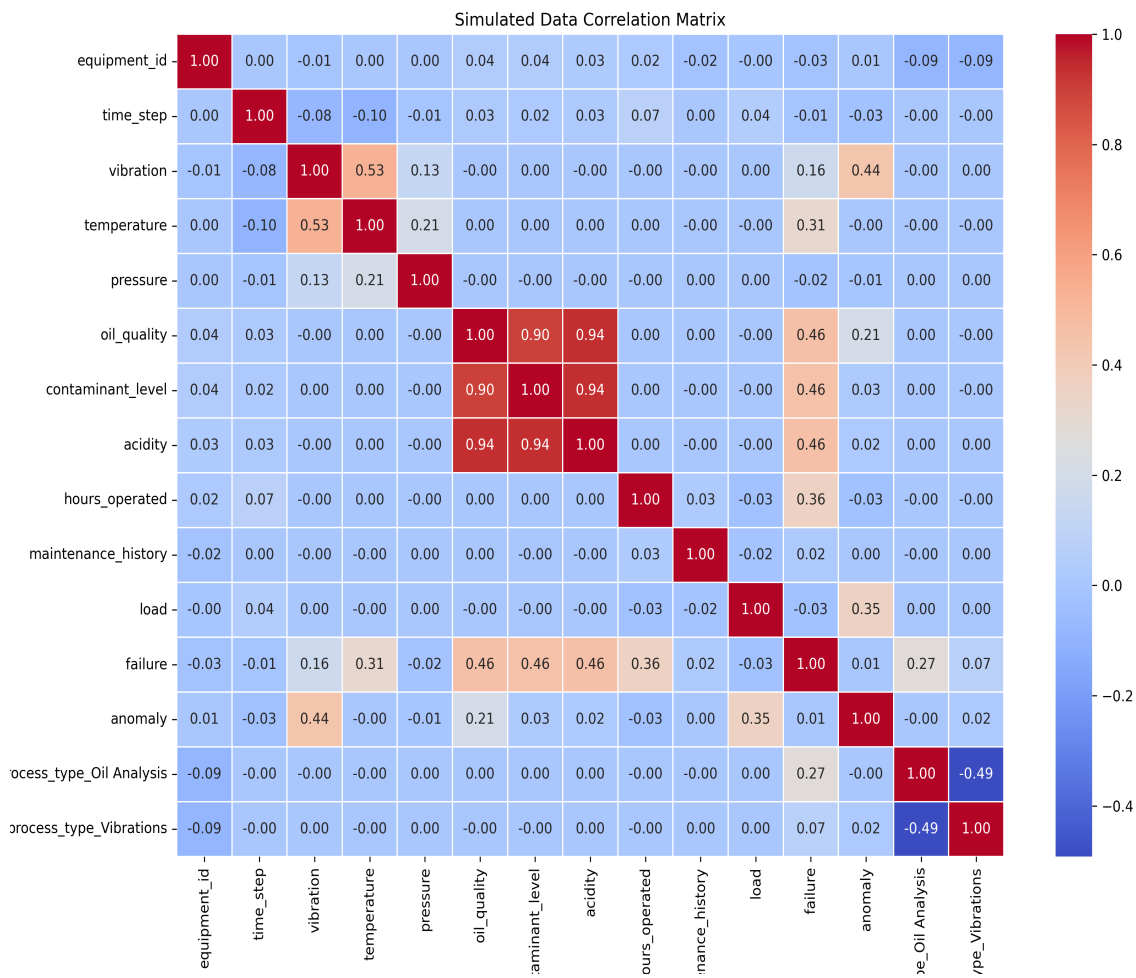


Figure 3: Variable Histograms

Histograms of Variables

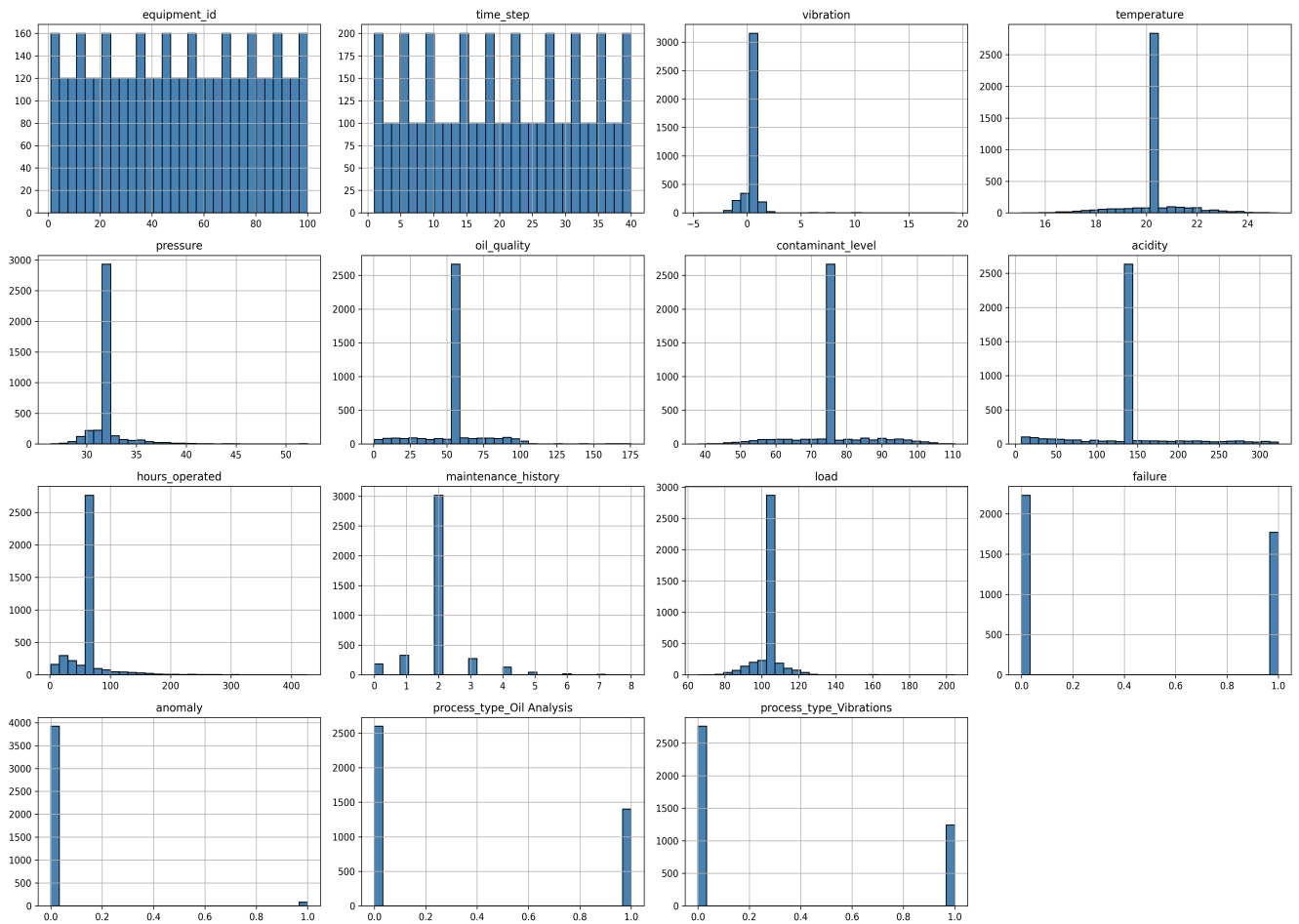


Figure 4: Variable Boxplots

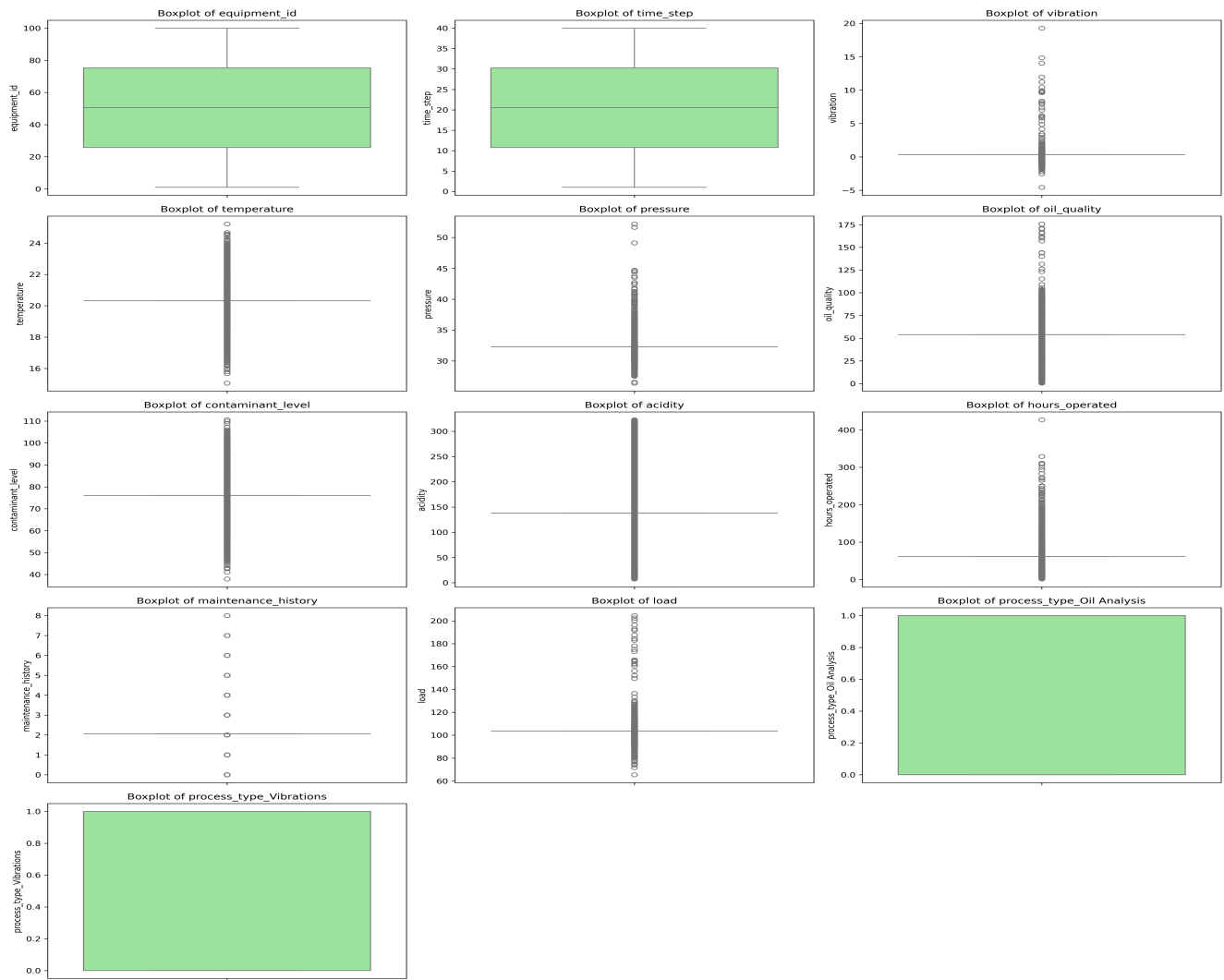
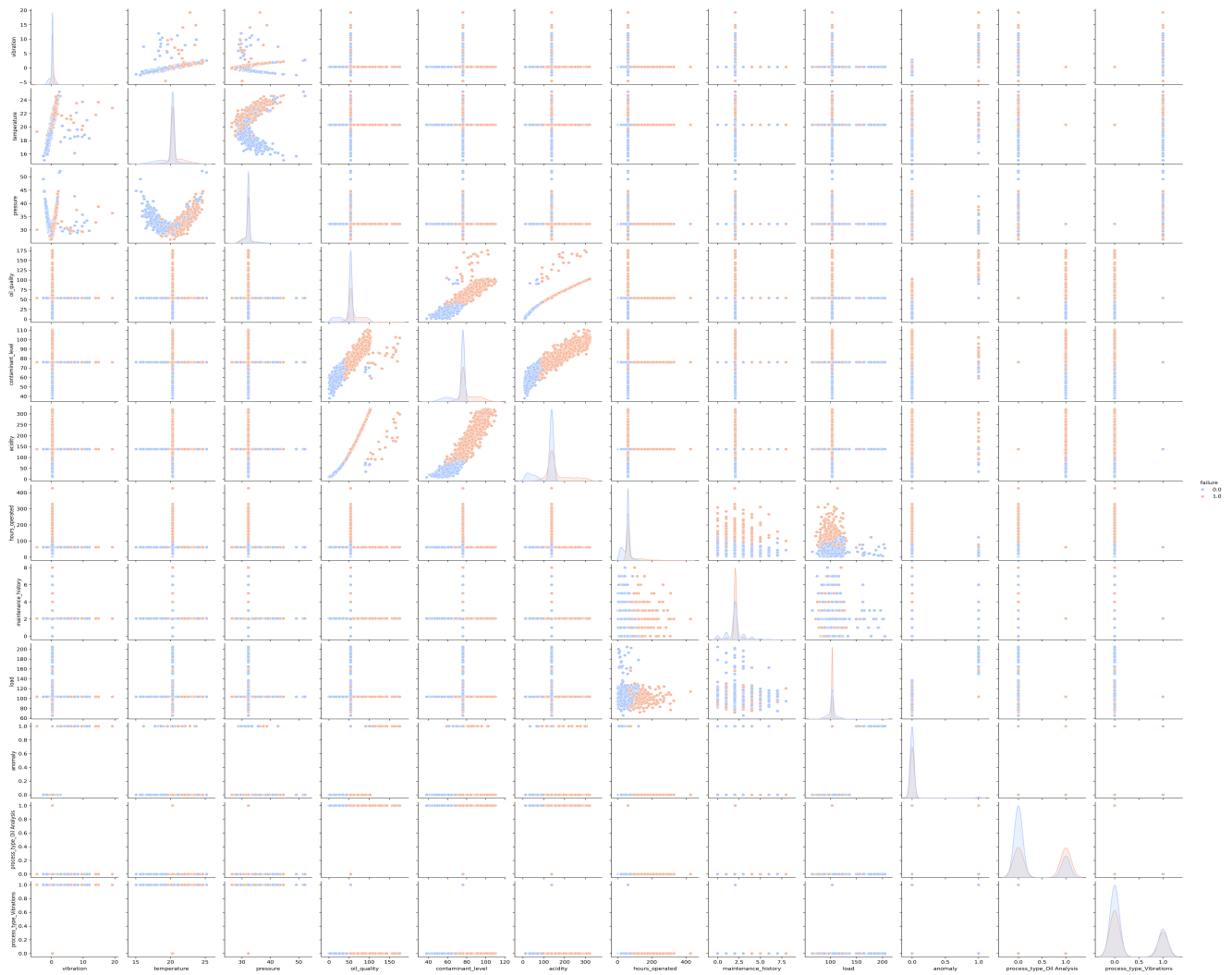


Figure 5: Variable Pairplot



2. Anomaly Detection

Five different anomaly detection algorithms were applied to identify unusual behaviors in temporal data.

Figure 6: Anomaly Detection - IsolationForest

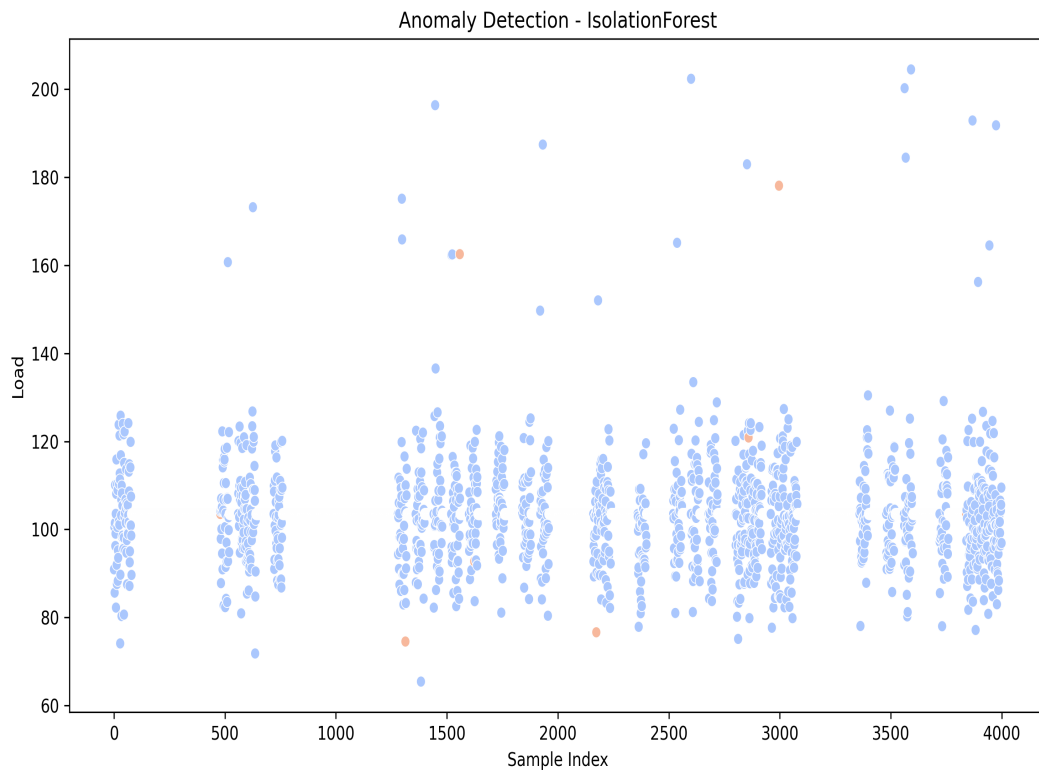


Figure 7: Anomaly Detection - OneClassSVM

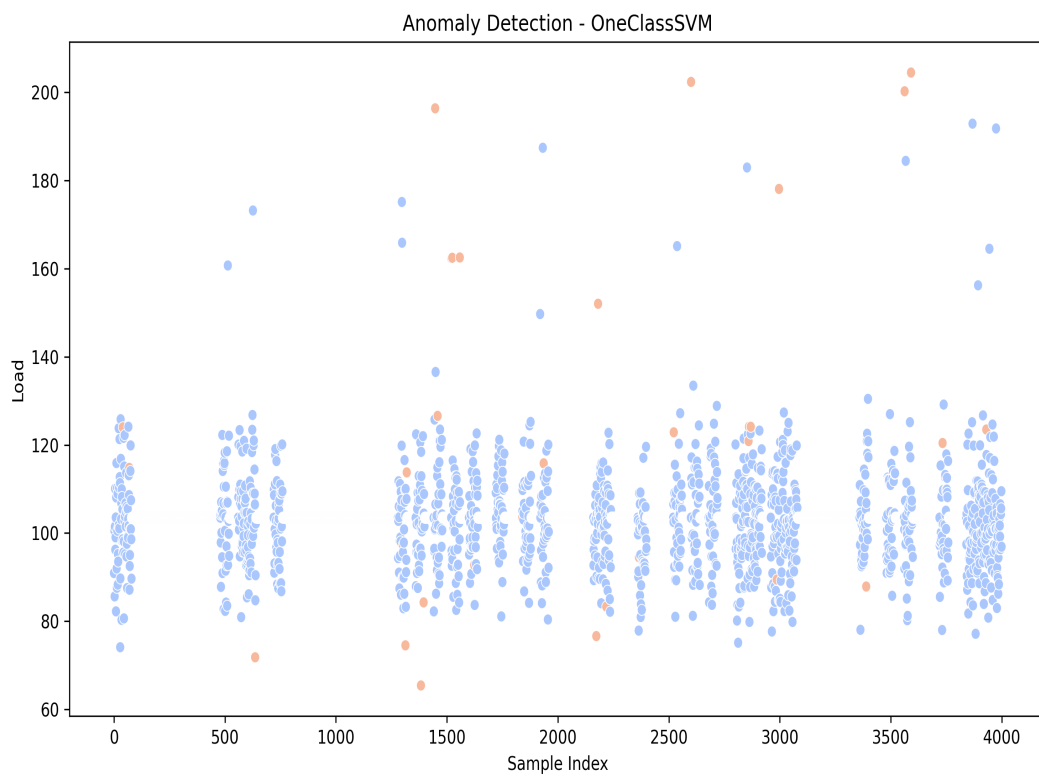


Figure 8: Anomaly Detection - LocalOutlierFactor

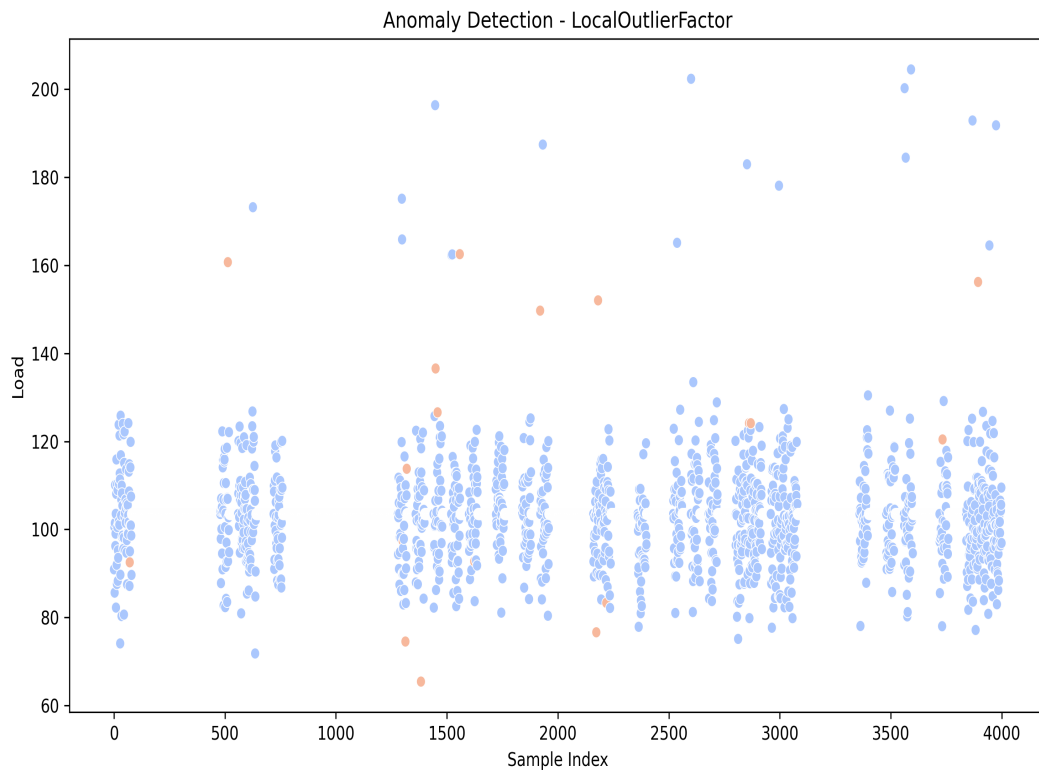


Figure 9: Anomaly Detection - DBSCAN

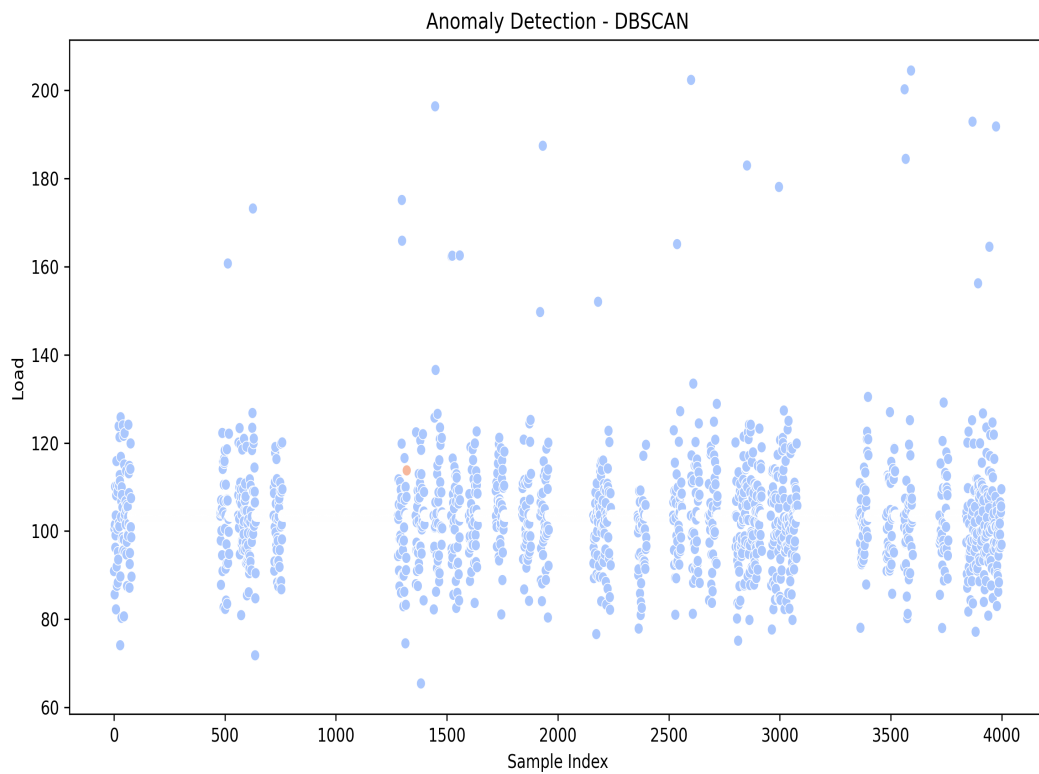
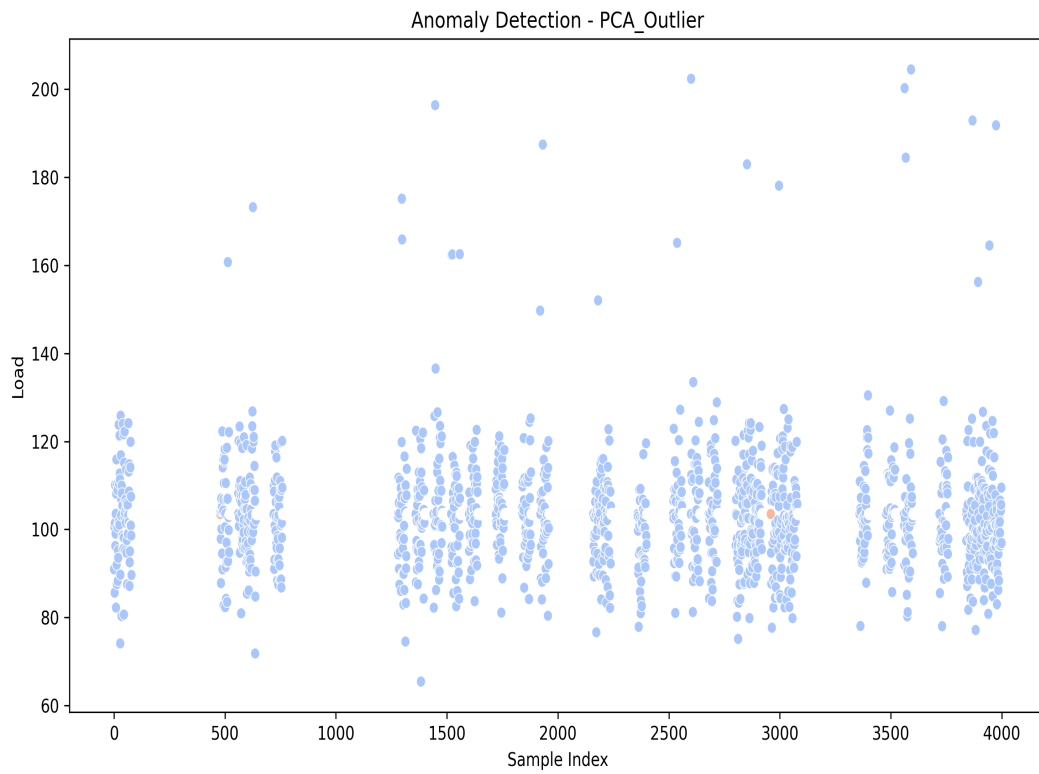


Figure 10: Anomaly Detection - PCA_Outlier



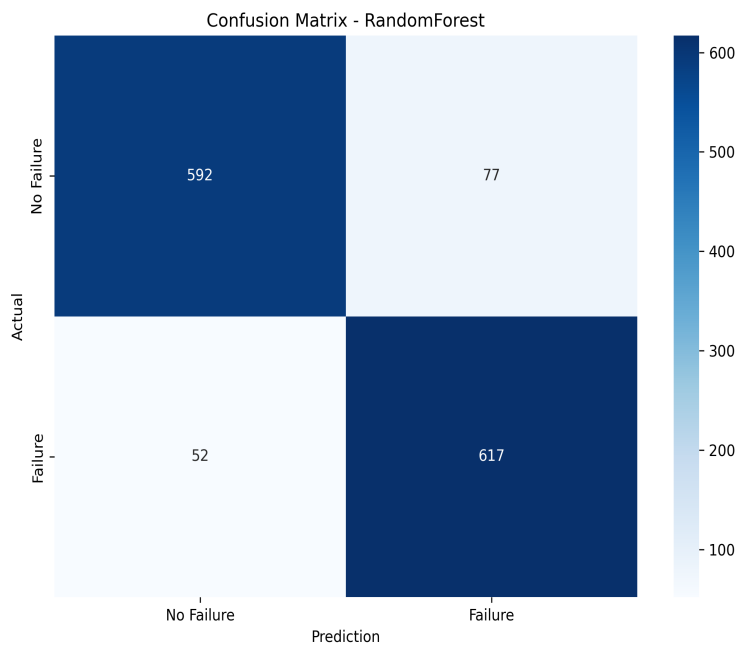
3. Classification Model Evaluation

3.1 RandomForest

ROC AUC: 0.9694

PR AUC: 0.9690

Figure 11: Confusion Matrix - RandomForest

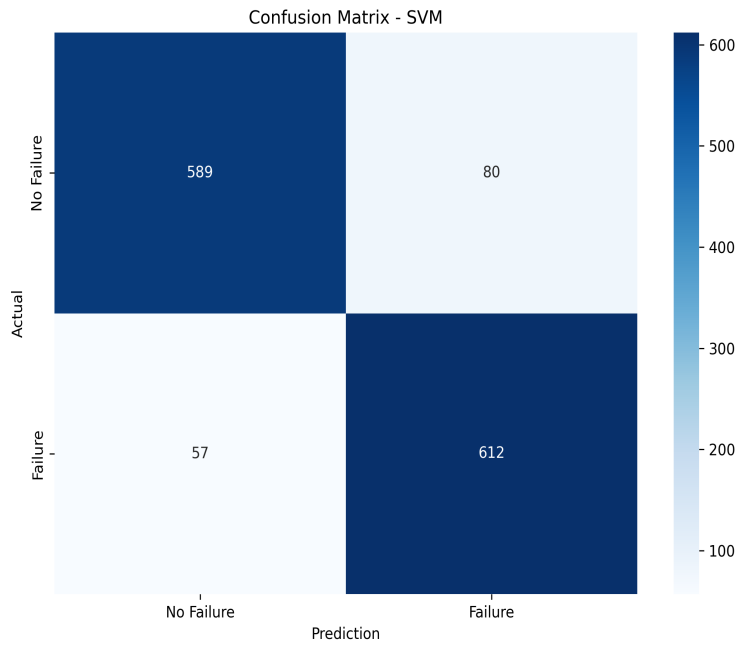


3.2 SVM

ROC AUC: 0.9683

PR AUC: 0.9683

Figure 12: Confusion Matrix - SVM

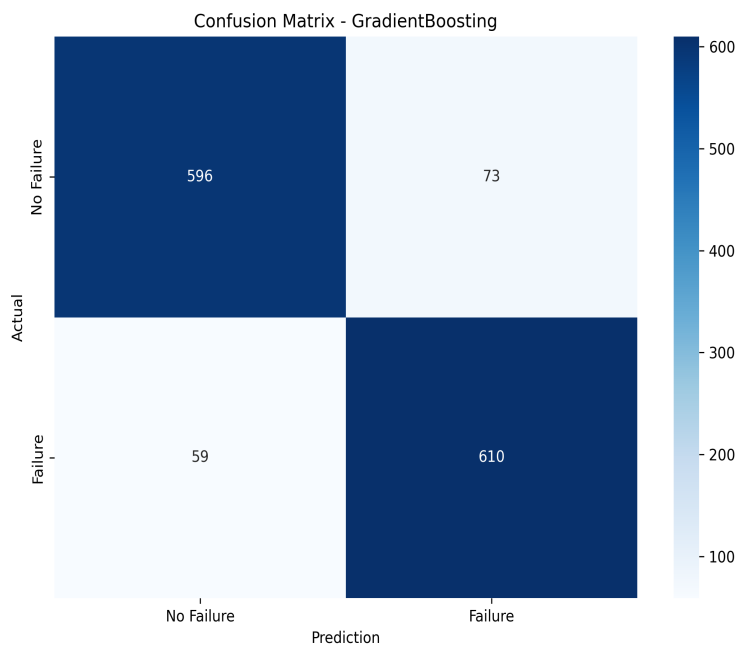


3.3 GradientBoosting

ROC AUC: 0.9679

PR AUC: 0.9672

Figure 13: Confusion Matrix - GradientBoosting



3.4 LogisticRegression

ROC AUC: 0.9670

PR AUC: 0.9668

Figure 14: Confusion Matrix - LogisticRegression

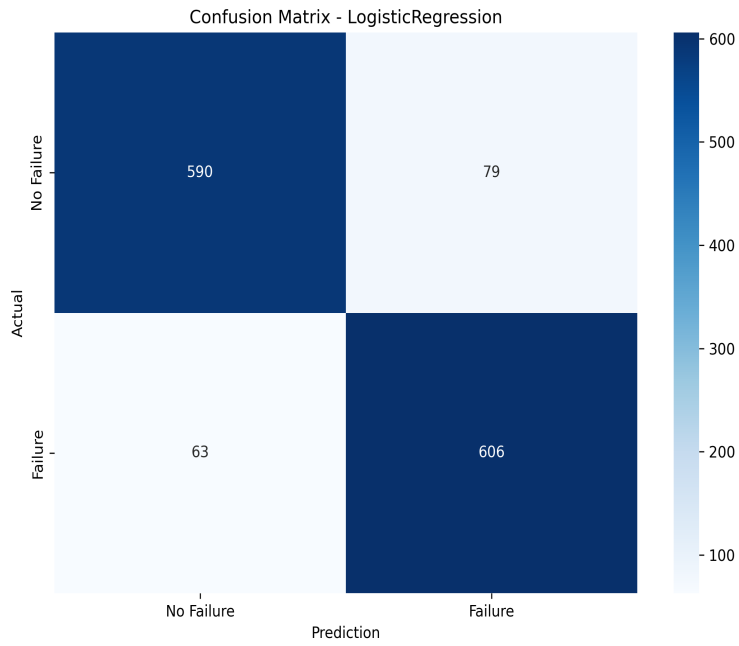
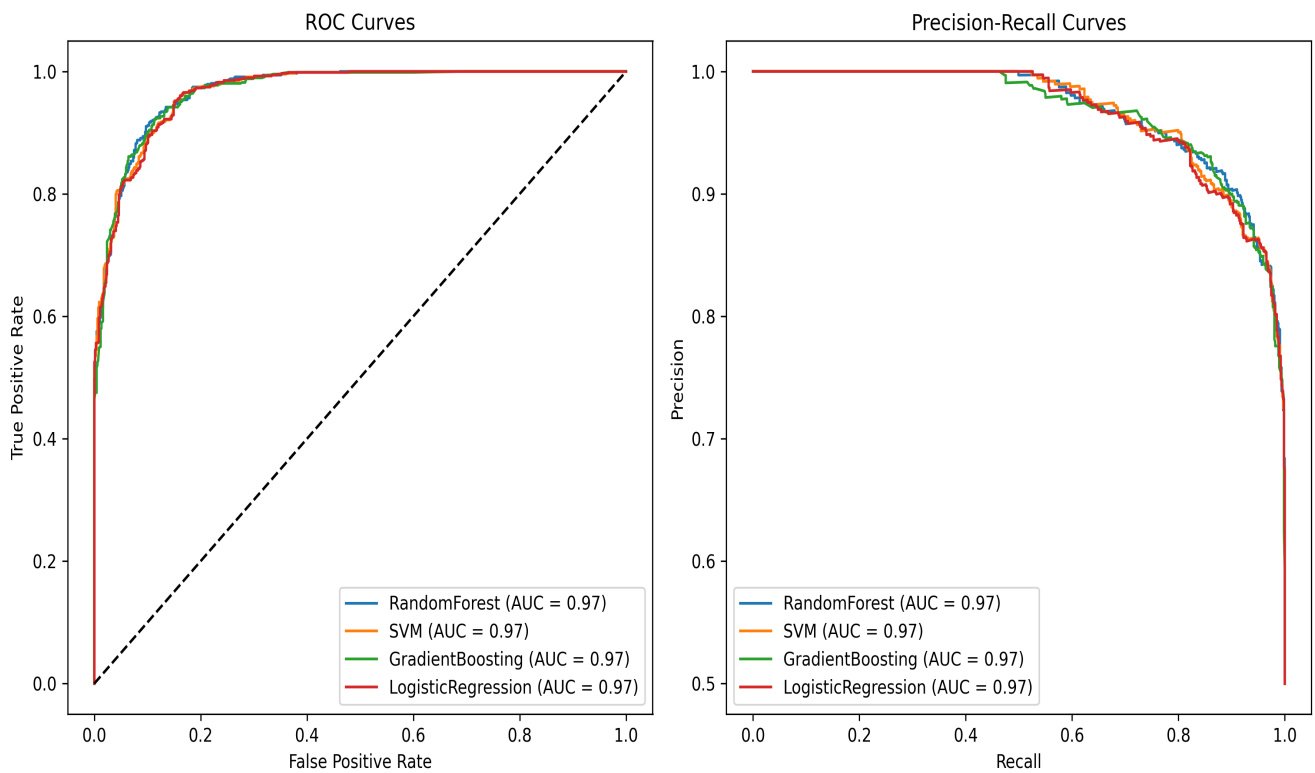


Figure 11: ROC and Precision-Recall Curves



4. Feature Importance

Figure 13: Feature Importance - RandomForest

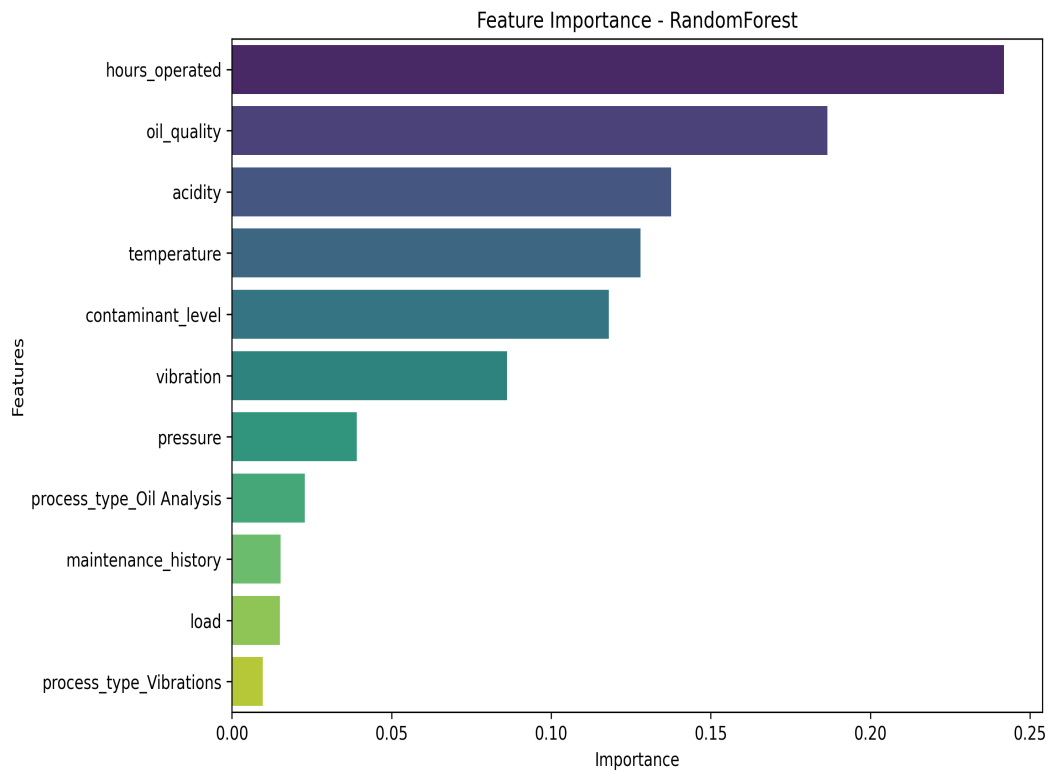
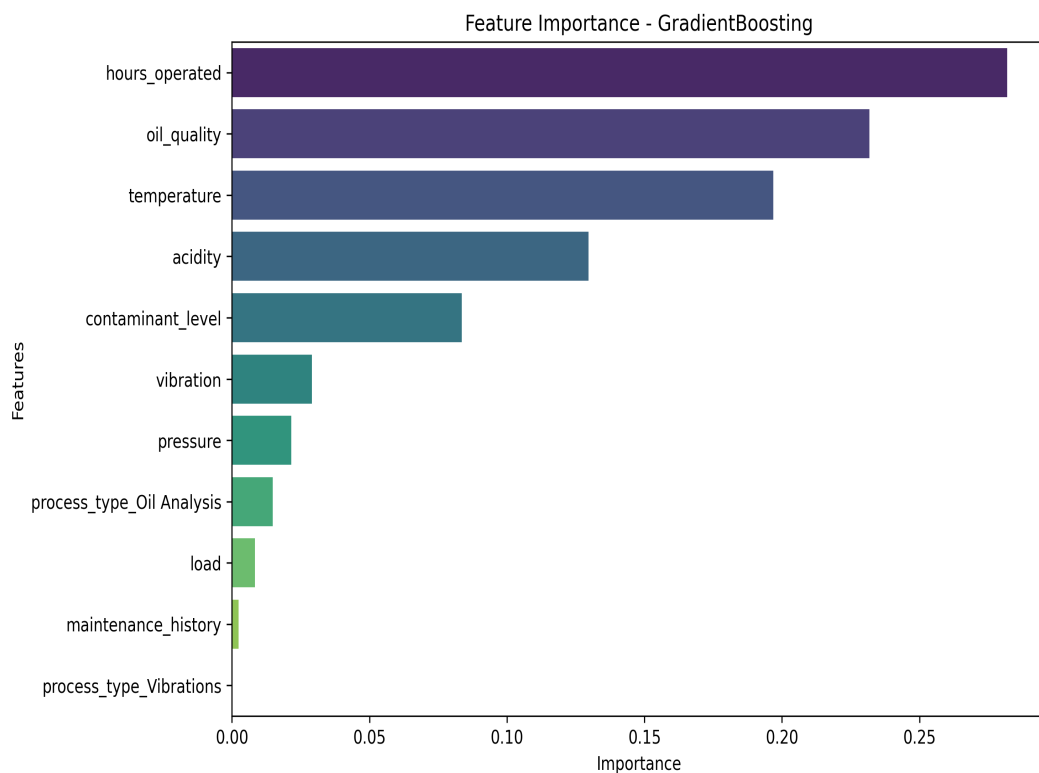


Figure 15: Feature Importance - GradientBoosting



5. Conclusions

The evaluated classification models demonstrated promising performance in predicting equipment failures. Among the evaluated models, Random Forest and Gradient Boosting showed the best performance metrics, indicating a high capacity to distinguish between equipment that will fail and those that will not. Anomaly detection using five different algorithms allowed identifying unusual behaviors in the temporal evolution of equipment. Variables such as vibration, oil quality, hours operated, and maintenance history were the most determining factors for predicting failures. These findings suggest that continuous monitoring and preventive maintenance based on these metrics can significantly improve equipment reliability.